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Journal:	*Supply Chain Management: an International Journal*
Manuscript ID	SCM-11-2023-0585.R2
Manuscript Type:	Original Manuscript
Keywords:	Case Studies, Corporate responsibility, Global Sourcing, Regulations, Responsible Trade, Sustainability

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Abstract

Purpose – Firms are increasingly pressured to comply with mandatory supply chain transparency (SCT) regulations. Drawing on information processing theory (IPT), the authors show how blockchain technology can address information uncertainty and equivocality in assuring regulatory compliance in an interorganizational network (ION).

Design/methodology/approach – IPT is applied in a single case study of an ION in the mining industry that aimed to implement blockchain to address mandatory SCT regulations. The authors build on a rich proprietary dataset consisting of interviews and substantial secondary material from actors along the supply chain.

Findings – Our case shows that blockchain *creates equality between actors*, *enables compliance*, and *enhances efficiency* in an ION, reducing information uncertainty and equivocality arising from conflict minerals regulation. The system promotes engagement and data sharing between parties, whilst protecting commercial sensitive information. The lack of central authority prevents larger partners from taking control. The system provides mineral provenance and a regulation-compliant record. System cost analysis shows that the system is efficient as it is inexpensive relative to volumes and values of metals transacted. Issues were identified related to collecting richer human rights data for assurance and compliance with due diligence regulations.

Originality/value – We provide some of the first evidence in the operations and supply chain management literature of the specific architecture, costs, and limitations of using blockchain for SCT. Using an information processing theory lens in an ION setting, we demonstrate how blockchain-based systems can address two key IPT challenges: environmental uncertainty and equivocality.

Keywords: *blockchain, conflict minerals, information processing theory, regulations, supply chain transparency*

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1. Introduction

"This is the future of mining and the value chains that involve it."

- Juan Luis Kruger, Minsur CEO

Information processing capabilities are critical to effective supply chain management (SCM) practices (Aben *et al.,* 2021). A substantial body of research in the operations and supply chain (OSCM) literature applies information processing theory (IPT) to investigate how such capabilities are used to overcome information challenges in supply chains. Information uncertainty and equivocality shortcomings have not been resolved in SCM, with dire human and environmental consequences (Jiang *et al.,* 2023), leading to regulation in which legislators mandate organizations to achieve supply chain transparency (SCT) of materials origin (Hofmann *et al.,* 2018). Little research exists that provides insights into SCM responses to the regulation or the information processing implications of delivering visibility and the required technological tools (Sodhi and Tang, 2019).

Digital technologies are employed to reduce uncertainty by enhancing information quality and quantity (Aben *et al.*, 2021) through improved information processing (Munir *et al.,* 2022) and analytical capabilities (Yang *et al.,* 2023). The availability of information between supply chain partners enhances communication (Faruquee *et al.,* 2021), facilitating visibility (Zhu *et al.*, 2018). Literature has investigated the impact of a variety of digital technologies (Garcia-Torres *et al.,* 2019), often dealing explicitly with high-level concepts such as 'digital technologies' (e.g. Aben *et al.,* 2021; Lorentz *et al.,* 2021) or 'supply chain analytics' (Srinivasan and Swink, 2018; Zhu *et al.,* 2018). Specificity of the technology employed, especially the role of blockchain, has received less attention, leaving open questions on how blockchain may be employed for SCT (Sodhi and Tang, 2019; Hastig and Sodhi, 2020).

Blockchain has emerged as a key underpinning technology for systems that promise SCT through traceability, providing a platform for collecting and sharing of information for assurance of materials provenance (Rogerson and Parry, 2020; Brau *et al.,* 2024). Blockchain technologies provide an interesting and potentially important subject for investigating the gap in the SCT literature at the intersection of SCT, technology, and interorganizational networks (IONs). Studies demonstrate blockchain's ability to trace various materials and products securely (Baharmand *et al.,* 2021; Danese *et al.,* 2021). However, cases in the published literature to date have investigated transparency for competitive advantage, meaning that information deemed unprofitable to collect can be omitted from the system. For example,

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Rogerson and Parry (2020) highlight a sustainable fishing case in which blockchain overcomes prior transparency issues by using satellite information to pinpoint vessel locations. That case shows that firms subsequently lose the transparency of the caught fish because, once a fish is processed, the different parts of the fish are more complex (i.e. less profitable) to track individually. In the case of mandatory SCT, such choices may lead to failure to comply with regulations because firms are required to track materials to their origin. The context for our study is the mining industry, an industry from which gathering provenance information "*can be very expensive and has questionable accuracy and reliability because of spot markets and co-mingled supplies*" (Gualandris *et al.,* 2015, p.7). Therefore, the mining industry and its customers must deal with issues relating to their inability to obtain the information they require and the reliability of the information they can find relating to the origin of materials. Activist organizations have sought to expose the damage done by some forms of mining to local environments and the human rights abuses, including child labour, associated with metal extraction supply chains (Gold *et al.,* 2015; Peng *et al.*, 2022). Supply chain issues have led to a rise in legislation over metals (Hofmann *et al.,* 2018), such as the European Union's (EU) new Conflict Minerals Regulation (European Commission, 2017).

As outsourcing is prevalent, individual focal firms often do not have access to information necessary to provide transparency of materials and operations (Meehan and Pinnington, 2021; Gualandris *et al.*, 2021). As a result, sustainable supply chain management increasingly relies on technologies to manage information flows (Alexander *et al.,* 2022; Corallo *et al.,* 2023) and ION collaboration to manage sustainability activities (Legenvre and Hameri, 2023). Blockchain-based systems have been proposed as a solution to providing necessary SCT (e.g. Kassis *et al.,* 2019; Nicola, 2019). However, little academic research examines how IONs may overcome the challenges of mandatory transparency of materials' origin. With (1) increasing regulation aimed at forcing more SCT; (2) the use of technology to help with regulatory compliance; and (3) interorganizational trends in overcoming intractable problems in supply chains, there is an important gap in scholarship at the confluence of these areas (Busse *et al.,* 2017). This study uses IPT (Galbraith, 1974) to investigate an ION's use of blockchain to comply with mandatory transparency regulation with the following research question: *How does blockchain address information uncertainty and equivocality in assuring regulatory compliance in an interorganizational network?*

Our unit of analysis is an ION that has deployed a blockchain system for regulatory SCT, created by blockchain provider firm Minespider; Minsur, a Peruvian mining company;

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and the multinationals Cisco, Google, and Volkswagen. The multinationals in the ION fall within the scope of the EU's Conflict Minerals Regulation, which came into force at the beginning of 2021 (European Commission, 2017). The ION therefore offers the advantage of enabling us to research current responses to regulation as they occur, including the governance mechanisms required for multiple multinational organizations to cooperate while enhancing information processing capabilities. Our setting is, therefore, unique since mandatory transparency is a novel regulatory mechanism. We use a dataset consisting of extensive interviews and secondary data in the form of company reports, industry press, white papers, and social media posts.

This study makes two contributions, which respond to calls (e.g. Fosso Wamba *et al.* 2021) for more research into how information uncertainties can be overcome using technology and empirical findings on blockchain (Hastig and Sodhi, 2020). First, we demonstrate that blockchain-based systems can help IONs overcome information uncertainty and information equivocality through information gathering and assurance activities. Our contribution builds on the work of Zhu *et al.* (2022) and Chittipaka *et al.* (2023), who find that blockchain can enhance performance and address some information challenges in supply chains by adding specific details to the data uncertainties.

Second, we contribute to phenomenological knowledge on blockchain in the OSCM literature and practitioners' understanding of the implications for blockchain's use for regulatory compliance. Our findings show that blockchain-based systems can enable compliance with conflict minerals regulation by providing evidence of materials provenance and that the scale required to benefit from blockchain (Sternberg *et al.*, 2021) can be overcome in part (but not entirely) in IONs. We detail the costs of setting up and maintaining such a system, which have not previously been seen in the literature.

The remainder of the paper is structured as follows. In section two, we review the relevant literature on SCT, IPT, and blockchain for SCM. Section three details the methodology, and section four demonstrates the findings. In section five, we discuss our findings in relation to the relevant literatures and managerial practice before concluding our study in section six.

2. Theoretical background

2.1 Supply chain transparency

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Transparency is an increasingly important aspect of SCM (Sodhi and Tang, 2019). Many drivers of SCT are voluntary, with resultant information retained by focal firms to reduce costs (Sodhi and Tang, 2019) and improve performance (Galati, 2022). Supply chains are under increasing regulatory and legislative scrutiny for their environmental and human impacts (Sodhi and Tang, 2019). To account for such external motivators, we adopt Baharmand *et al.'s* (2021, p.1523) definition of SCT: "*reporting to and communicating with stakeholders to provide traceability regarding the history of the products and visibility about current activities throughout the supply chain*." Baharmand *et al.'s* (2021) definition is distinct from other attempts to clarify the meaning of SCT because of their reference to 'stakeholders', such as regulators and customers - see, for example, Rogerson and Parry (2020), who focus on visibility under the term 'transparency', and Sodhi and Tang 2019, who differentiate between the internal focus of visibility versus the external focus of transparency. The importance of SCT for wider supply chain stakeholders has increased due to prior supply chain failures, such as the Rana Plaza disaster (Garcia-Torres *et al.,* 2019), the COVID-19 pandemic (Uddin *et al.,* 2023), and wide-ranging sustainability challenges (Hastig and Sodhi, 2020).

SCT can be difficult to deliver in some inherently complex supply chains (Swift *et al.,* 2019; Khaoua *et al.,* 2022) arising from the number of suppliers, their geographical dispersion, and the quality of their interrelationships (Kim and Davis, 2016). The components for modern goods are frequently sourced through a supply ecosystem rather than a traditionally conceptualized linear supply chain (Chowdhury *et al.,* 2023). The resulting opaque, multi-layered system presents difficulties in delivering transparency, creating substantial reputational risks for focal firms (Wilhelm *et al*., 2016). Environmental and human abuses arise and persist, in part, from continued supply chain opacity (Soundararajan *et al*., 2021), which has had two important results relevant to our study. First, firms increasingly seek to manage sustainability risks (and other complex issues, Roehrich *et al.,* 2023) through multi-organization collaborations known as IONs (Daghar *et al.*, 2021). Second, markets' inability to solve intractable environmental and human rights abuses in supply chains has led to an increasing volume of legislation mandating disclosure and, more recently, action (Cole and Shirgholami, 2022).

Legislation in the USA, France, and the UK mandates firm disclosure of policies and practices related to modern slavery and human rights (Rogerson *et al*., 2020). Legislation in Europe is forcing firms to take action to identify, mitigate, and remediate environmental and human rights abuses and then to report transparently on that process (Schrempf-Stirling *et al*.,

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2022). Poor industry practice has led the USA and EU to create legislation targeting minerals or metals from conflict-affected and high-risk areas, the so-called '3TG' metals: tin, tantalum, tungsten, and gold. From January 2021, all firms importing such minerals into the EU must provide evidence of provenance (European Commission, 2017), mandating all firms operating in the world's largest trading bloc trace materials and products upstream in supply chains. The legislation is founded on the principle of transparency through reporting (Rogerson *et al*., 2020), and the traceability aspect of transparency is becoming core to efforts by firms (Cousins *et al*., 2019) and legislators (Cole and Shirgholami, 2022). Traceability adds to transparency by providing data on materials and goods flow from point of origin. Traceability captures information about each hand-off along the supply chain to ensure quality, counterfeit detection, and social and environmental factors (Rogerson and Parry, 2020).

Systems and processes to provide traceability have been difficult to develop because they rely on collecting, collating, and analyzing information from multiple firms in supply networks (Cousins *et al*., 2019). Achieving transparency in metals supply chains had appeared an intractable problem. For over a decade, the Responsible Minerals Initiative (RMI) has attempted to embed its standards and those of the OECD due diligence guidance (OECD, 2016), including environmental and human rights assurance, into benchmarked practice. However, creating reliable supply chain traceability of materials has thus far proven problematic (Hastig and Sodhi, 2020). Mass balance, a system where the mass of certified outputs sold cannot exceed the mass of certifiable inputs adjusted for process yield, is used extensively in commodities. The approach allows certified and uncertified materials to be mixed within a process. Where materials are mixed, specific materials cannot be traced beyond the site from which they were removed. All outputs are effectively 'contaminated,' removing transparency and creating information uncertainties for firms trying to trace materials (Lafargue *et al*., 2021).

2.2 Information processing theory

The difficulty of generating and sharing trustworthy information creates uncertainty, limiting firms' ability to assure sustainability in their supply chains (Giri and Manohar, 2023). As information management has become increasingly important in OSCM, IPT has become a relevant research lens (Aben *et al.,* 2021; Giri and Manohar, 2023; Laari *et al.,* 2023). IPT is concerned with how firms manage the interconnectedness of uncertainty, and their information processing needs to achieve a given level of performance (Galbraith, 1974). Information

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processing encompasses the "*gathering, interpreting and synthesis of information in the context of organizational decision making*" (Tushman and Nadler, 1978, p. 614), which is required to match uncertainties. A good "fit" of information processing needs, with limited uncertainty, improves a firm's performance, whereas a lack of congruence leads to decreased performance (Tushman and Nadler, 1978).

Two types of uncertainties are identified in IPT, namely, uncertainty arising from the lack of information (environmental uncertainty) and information ambiguity (equivocality) (Aben *et al.,* 2021). These uncertainties alter information processing needs, which must be managed through adequate management practices. As uncertainty increases, firms may reduce the information processed by creating slack resources (e.g. a buffer of inventory), establishing self-contained tasks (e.g. setting up project groups), increasing information processing capacities (e.g. joint decision-making processes across lines of authority), or implementing information systems (e.g. human-machine combinations, computers, IT) (Galbraith, 1974).

IPT studies in the OSCM literature offer several discrete strategies for managing uncertainties, building on Galbraith's (1974) conceptualizations. Research finds that interorganizational work is key to such efforts (Aben *et al.,* 2021). Two notable strategies are to improve the quantity and quality of information available to build long-term collaborations with partners (Laari *et al.,* 2023) or to deploy digital technologies (Aben *et al.,* 2021). Supply chain complexity means that, in practice, these strategies are best deployed simultaneously. Technology alone cannot reduce equivocality (Rogerson and Parry, 2020), and over-reliance on technology can hinder coordination, reducing information processing capacity (Yang *et al.,* 2023).

Supply chains in the mining industry face particular challenges with information processing capabilities (Srinivasan and Swink, 2019). Metal ores often come from remote areas, many sources may be aggregated together and traded as commodities (Kim and Davis, 2016). The opacity of minerals supply chains means that obtaining information from mines and the communities in which they are situated is difficult. Even where information has been obtained, verifying its reliability is challenging (Hofmann *et al.,* 2018). SCT therefore remains an important problem that IPT scholars have only recently begun to investigate, with limited in-depth case literature, especially from an interorganizational perspective (Zhu *et al.,* 2018). Our case study seeks to address this important gap in the literature on information processing in the context of the mining industry that struggles to obtain reliable information.

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2.3. Blockchain technology in supply chain management

Data storage decentralization, a core design feature of blockchain, and the cryptographic security and information transparency the technology offers, make blockchain an effective tool to store and share tamper-proof data (Brookbanks and Parry, 2022) if the governance of the system can be aligned with the use case (Goldsby and Hanisch, 2022). Managing access to blockchain data is achieved through 'permissioning'. Access can be given to everyone (a public blockchain), limited to a specific group (a permissioned blockchain), or limited to a single individual or firm (a private blockchain) (Rogerson and Parry, 2020). Traditional information storage systems use centralized servers that enable the owner to control information. Such systems are sufficient if the information controller is a trusted party (Cole *et al.,* 2019), but where no trusted central authority exists, blockchain allows groups to share a common dataset.

Blockchain's suitability for SCT is well documented (Shahzad *et al.*, 2024). Numerous examples of blockchain use cases of SCT for provenance of materials and goods have been provided by industry (Deloitte, 2017; RCS Global, 2018; Norton and Conlon, 2019), but few empirical scholarly studies exist in leading academic journals (Sternberg *et al.,* 2021). Little is known about how blockchain systems are deployed, their architectures, specific features, the extent to which they solve the problems they claim to address, or their cost (Cole *et al.,* 2019). Little attention has been paid to IONs' capacity to collect and process information via blockchain applications (Hastig and Sodhi, 2020; Galati, 2022). The research gap on blockchain in IONs is particularly acute given that published research posits that scale (van Hoek, 2020), which cannot be delivered through private blockchains (Baharmand *et al.,* 2021), is required to benefit from the technology fully. Extant research has been criticized, being called "atheoretical", rooted as it is in individual applications of the technology (Sternberg *et al.,* 2021). Scholarly and industry knowledge of blockchain's benefits (and limitations) compared to other tools and approaches to information processing in supply chains is therefore limited (Zhu *et al.,* 2022).

An important gap in the literature exists at the intersection of SCT, IPT, and conflict minerals. Blockchain solutions have been proposed as a solution to the challenges in the area, and the study of these systems in context contributes to knowledge of how information processing capabilities can be enhanced to reduce uncertainties and enable firms to comply with SCT regulations. This gives rise to our research question: *How does blockchain address*

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information uncertainty and equivocality in assuring regulatory compliance in an interorganizational network?

3. Methods

3.1 Research approach, sampling and setting

To answer the research question, we focus on a single exploratory case of blockchain integration for mandatory SCT. The single case study is appropriate for studying a complex phenomenon in its natural setting and deriving an explanation in greater depth (Voss *et al.,* 2002). Case studies are used for their ability to answer 'how?' questions (Voss *et al.,* 2002) and to examine current issues and responses in a specific context (Siggelkow, 2007). Therefore, the approach is well-aligned with this study's objectives of responding to calls in the literature for more empirical work on current issues around SCT (e.g. Sodhi and Tang, 2019), and the use of blockchain in SCM (Hastig and Sodhi, 2020) to achieve regulatory compliance. Abductive reasoning was appropriate in this study as it allowed the authors to move between concepts and empirical data. The unit of analysis is an ION using blockchain for mandatory SCT. The methodological framework for this study is presented in Table I.

< Please insert Table I about here >

To address the research objectives, we sampled based on two criteria. First, we sought examples of organizations using blockchain to help with regulation. We identified several sectors (e.g. mining, aerospace, food) where blockchain provided a solution to information processing for SCT. The mining industry was identified as a sector confronted with increasing regulation regarding conflict minerals. SCT was important to the mining industry, which is known to harbour unsustainable practices (Srinivasan and Swink, 2018), including environmental degradation, child/forced labour and poor working conditions. Increasing demand is likely to amplify further pre-existing unsustainable practices (Hofmann *et al.*, 2018). Second, we sought organizations that were part of IONs where the characteristics of blockchain systems played a role in providing data integrity between parties. We sought cases where we could examine how IONs implement blockchain technology to address information uncertainty and information equivocality in response to mandatory SCT demands. Mining industry supply chains are opaque, consisting of multiple actors, many based in remote parts of the world (Kim

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and Davis, 2016). Downstream focal firms often rely on data from their IONs for compliance reporting (Srinivasan and Swink, 2018).

Recruiting project partners who were actors in an ION in the mining industry was challenging. Four mining industry blockchain service providers were invited to participate in this research project. Only one service provider, Minespider, offered us the access required to conduct an in-depth case study. Minespider is a European technology provider founded in response to responsible sourcing concerns in global supply chains. They are the technology provider to an ION using blockchain for compliance, which had recently won EU funding. The consortium of firms, including Google, Volkswagen, and Cisco, sources cassiterite, the ore from which tin is produced (Härkönen, 2018), from Minsur's San Rafael mine in Peru. As with most metal ore, cassiterite is a commodity traded on global markets, and may be sourced from various locations. The ore is mixed and smelted to create tin, which is used extensively as solder in electronics. Manufacturers and focal firms find it difficult to trace the origins of tin as provenance is lost at multiple nodes of the supply network (Busse *et al.,* 2017).

The immediate driver for the Minespider ION's adoption of blockchain was the enactment of the EU's Conflict Minerals Regulation, which came into force in January 2021 (European Commission, 2017). The decision to trace tin from Peru was a result of Minsur's membership of the International Council on Mining and Metals, which has a strong focus on sustainability, and the company's prior sustainability work. The case is significant to the market as operations at San Rafael account for 12% of global tin production [archival document A], making the project a large enough context in which to conduct an in-depth case study.

3.2 Data collection and sources

Data collection began in January 2019 and ended in April 2022. We collected primary data (interviews) and secondary data (e.g. firm disclosures, industry reports, social media posts) (see Appendices A, B, C, and D). A five-step data collection process was employed. First, secondary data were collected to provide initial insights into SCT challenges in the mining industry, how the selected firms communicated the use of blockchain to address these challenges and to understand the regulatory demands of governments, NGOs, and consumers for SCT.

Second, we conducted two exploratory interviews with the technology provider (Minespider). The initial discussion was conducted to confirm the appropriateness of the case and gain initial insights into the ION members that used blockchain to meet regulatory demands

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for SCT. The initial interviews enabled us to refine the interview protocol (Appendix E) and apply snowball sampling, requesting contact information for potential interviewees who could help us answer the research question.

Third, 13 semi-structured interviews were undertaken, lasting between twenty-seven minutes to two-and-a-half hours. Initial interviews were conducted with Senior Manager 1, Senior Manager 2, and an Innovation Manager, with further snowball sampling requests. The authors were given a demonstration of the tool and its functions by the project manager responsible for software for the tin project, and a technology lead clarified further system points. All interviews were conducted via telephone and video conferencing software by one author.

Fourth, secondary data was collected about the IONs, consisting of industry reports, company website data, press releases, and media articles found both directly on mining industry websites and through social media posts on Twitter, Facebook, and LinkedIn, and YouTube (Longoni *et al.,* 2019; Bansal *et al.,* 2020). We used a strategy of two authors coding our data, follow-up interviews for cross-questioning respondents to clarify apparent inconsistencies, and triangulation of primary and multiple sources of secondary data and prior findings in the literature to ensure the validity of our findings (Gibbert *et al*., 2008). Step five involved email exchanges and face-to-face meetings over a two-year period with the interviewees for clarification of specific points, understanding of apparent disparities between secondary data and primary data, technology updates, and presentation and verification of the findings and conclusions of this final paper.

In total, over 22 hours of interviews were conducted, 42 emails exchanged, 1.15 hours of video material evaluated, and 390 pages of documents analyzed. The list of data sources and our interview protocol are presented in Appendix A.

3.3 Data analysis

We conducted the data collection and analysis in parallel using the NVIVO data analysis software. The data analysis followed the open, axial, and selective coding procedures after Strauss and Corbin (1990). Open coding involved analyzing all the interview and secondary data. Therefore, a line-by-line coding strategy was used motivated by the gaps in the literature, with open codes around SCT, collaboration of IONs for SCT, and the efficacy of blockchain for compliance among IONs. To ensure the quality of the initial codes, the principal researcher

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and one co-author analyzed and discussed the resultant coding for comparability, thus ensuring intercoder reliability. The focus of analysis on issues related to IPT meant there were few discrepancies between the codes the two authors produced. Instances of discrepancies related to interpretations of the literature and were resolved by discussion between all authors.

In the second phase, core interview data were analyzed using the same line-by-line technique to decompose data into more concrete themes. The principal investigator and one co-author read the data, comparing the emergent themes based on primary and secondary data with the pre-existing literature on SCT, IPT, supply chain regulation compliance, and blockchain in SCM. In this way, we built the case structure, identifying the major points emerging from data, which were central to the case and important in answering our research question. Inconsistencies in case understanding required us to seek clarification from interviewees and search for further data sources. Initial codes referenced the interviewees' exact; for example, blockchain *"eases interorganizational data sharing"* or *"avoids monopolistic tendencies of large blockchain owners."*

The third step involved systematically comparing different initial codes and group emergent codes into related concepts. The coding process at this stage was iterative (Corbin and Strauss, 2008). The principal researcher and one co-author cross-checked the validity and credibility of codes that were combined into categories. For example, *"provides a traceable chain of information for provenance"* and *"enables compliance with regulations in the EU and USA"* were grouped under blockchain *"offers a regulation compliant record traceable to provenance."*

The final step in the coding process involved selective coding by consolidating emergent themes into core categories. For example, "*offers a regulation compliant record traceable to provenance*", *"offers site-specificity for audit but does not replace audit process"*, and "*increasing institutional demand for responsible mineral sourcing*" were grouped under the core category "*enables compliance*". The result of our coding process can be seen in Figure I below.

< Please insert Figure I about here >

4. Findings

The following section highlights the themes identified in the analysis, provides a table of quotations which support that analysis, and outlines the relationship between the themes within

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our findings. As shown in Figure I, we identified three themes in our data related to the benefits to IONs of using blockchain to comply with conflict minerals regulation. Our themes are *Creates equality between actors*, *Enables compliance*, and *Enhances efficiency*. The primary and associated second-order themes can be seen in Table II, below. The following sections offer more detail on each of the second-order themes of our first theme, *Enables compliance.*

< Please insert Table II about here >

4.1 Creates equality between actors

Our first theme consists of the second-order codes *Provides a platform for trust between parties sharing a supply chain* and *Democratizes information sharing to reduce tensions and enhance scalability*. These themes speak to how the governance of the blockchain system promotes engagement while protecting commercially sensitive information.

Provides a platform for trust between parties sharing a supply chain

Our data show the strategic choices made regarding blockchain system structure, accessibility, and points of centralization. Minespider's responsible sourcing solution employs a proprietary blockchain system built on top of the public Ethereum blockchain. The choice of a public blockchain was taken to give more flexibility and scalability to the system if demand grew. A senior manager told us that "*this single system allows any number of responsible suppliers to join, creating a web of custody, as opposed to only an end-to-end chain of custody from mine to end user*" [respondent 1].

Minespider's solution for tin mining supply chains streamlines the process of compliance, offering assurance of provenance by enabling the sharing of verifiable information. The system is built on the philosophy that "*the best use of blockchains is in securing trust among firms that didn't necessarily trust each other*" [respondent 1]. A senior manager explained that capturing the identity of individuals entering data on the blockchain drives behaviours, such as accuracy in data provision, that help engender trust "*by putting more pressure on the person who enters the data. When you know the data goes onto a blockchain, you behave differently [...] In the physical world, we can do more and better because we can check at any time after the fact*" [respondent 2].

The technology readily scales as a large number of nodes can be added to the system. The major partners' commitment to responsible supply chain management enables the case study example. Scaling beyond one mine or introducing many more powerful firms into the

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ION increases the potential for conflicting priorities creating division. Even in the current case, it was recognized that asking for too much data from partners might have adverse effects on system performance: *"Although it is tempting to consider all the different data points that can be entered, consider sending questionnaires that are no more than 3-5 questions to suppliers... It is important not to overburden participants with information requests"* [archival document G].

Democratizes information sharing to reduce tensions and enhance scalability

The Minespider system is based on three 'nested' levels of privacy. "*These layers divide the data into a private layer, offering certainty for regulation; a transparency layer, sharing data downstream for certifications and provenance, but not competitors or the public; and a public layer available for anyone to see. The key to this system is that the encryption is done by the data owner, meaning that they alone have the power over those data. This is how our blockchain is decentralized, removing the need for a central authority holding passwords. It's a system designed without a central authority*" [archival document I]. The three-layered system ensures that commercially sensitive information remains accessible only by the owner of that information. Therefore, security for each party is assured, retaining information that firms do not want to share and ensuring that there is no illegal sharing of information that might subject firms to anti-trust investigation. *"It is this nested technology that makes it so that you are in total control of your data, but there is no bank, or no central authority, or no central company, or government that has backdoor access to it"* [respondent 1].

The decentralized system is also designed such that *"it runs without Minespider. We add nodes when needed, but the system runs without us"* [respondent 2]. System design is important for both firms and the future of blockchain-empowered metals traceability because the decentralization of the system and the lack of centralized control by any organization act as a check on domination by one or a few parties. Google reports that Minespider's approach "*avoids the risk of a large player with a private blockchain creating a monopoly on the global mineral supply chain. Minespider's design provides a way to decentralize protocol governance while preserving the privacy of proprietary data*" [archival document A].

Access to information for relevant nodes in the supply chain is provided through the 'digital product passport' (DPP). Users can scan a QR code to see the shipment linked to that code and its related materials provenance, compliance information, carbon emissions, and recycling information on a mobile phone screen. The system design enables firms to retain

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control of own information and grant permission to other parties when needed, furthering the decentralization of the system. The system's availability provides key information that allows each firm to communicate relevant sustainability information to stakeholders.

4.2 Enables compliance

Increasing institutional pressure for responsible minerals sourcing

Statements from respondents identified the increasing demand for information about the origin of goods and materials that underpins Minespider's work at San Rafael. This motivation is an important factor because of the differences between the information required for regulatory compliance and the broader information demanded by stakeholders. A senior manager told us that: "*Knowing where your metals come from that go into electronics is becoming normal, expected*" [respondent 2], which acts as a driver for the implementation of systems to deliver efficient and effective information flows.

Offers a regulation-compliant record

Cassiterite ore is unique in its geolocation, allowing mined material to be identified by its chemical composition. Each batch of cassiterite sample is geochemically tested for these specific data points and for quality by a third-party laboratory. Minespider uses this information to create 'digital product passports' (DPPs)[1], described as "*a blockchain-secured digital identification for a material shipment. It contains all the information needed to cross a border, comply with due diligence requirements, and recycle your minerals*" [respondent 2]. Once digitized, the information is used to create a digital asset on the blockchain, and this digital-to-physical-asset link is secured at the source and verifiable later. A project manager explained that "*as tin is mined, a product passport is created for that tin, which is combined with other product passported tin until a shipping container of tin is available, at which point a new passport is created, subsuming the prior passports within it*" [respondent 4]. The DPP is created using a web page, so no app needs to be downloaded for the purpose. Each specific shipment is tested and tagged using a QR code. A project manager explained that a key challenge in sourcing responsible tin is caused by the metal being traded mass balance. Mixing tin ore from other sources jeopardizes the traceability of the metal, since its chemical 'fingerprint' is altered. San Rafael produces enough cassiterite continuously to support the

[1] Digital product passports are not essential for tracking minerals but the European Commission has proposed DPPs as a tool for traceability, especially into circular economy models. https://psqr.eu/publications-resources/all-about-dpp/

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process requirements of the supply chain partners. Being able to use one source of tin enhances traceability and the system's efficiency at scale. No time is lost waiting to make up the correct size of ore shipments, and chemical composition identity is maintained facilitating tracing throughout the supply chain.

The materials' DPP and logistics data create a chain of custody, providing traceability along the supply chain and facilitating the recycling of tin not used in production. Establishing a chain of custody "*gives companies insight into the origin of the metals and the conditions under which they have been produced*" [respondent 1]. Chain of custody for materials is essential in creating rigorous human rights audits of the labour employed in the supply chain. Therefore, Volkswagen uses the Minespider blockchain system to deliver on its obligations, using the data for "*eliminating sources of error, guaranteeing social and ecological standards*" [archival document F]. DPP enables circular supply chain traceability, but the validity of the claims over materials is compromised if material from a different unaudited mine source is added.

Does not replace prior audit processes

The physical-digital link between activities and assets and their digital records remains imperfect because the San Rafael project currently relies on RMI audits, with certification saved onto the blockchain. Information collected in and around mines regarding human rights conditions and regulatory violations is important, but information collected cannot easily be verified. While IoT devices and sensors can capture environmental data, other *"data points are collected manually"* [archival document G]. Having a digital representation of a site (of production, for example, which does not change) is much simpler than creating that physical-digital link for conditions at that site, which are less stable when studied in detail. Conditions are dynamic and social audit information is qualitative in nature, difficult to measure, and transitory (e.g. working conditions, work hours, pay, freedom of association, freedom to choose work, freedom to leave). Human rights compliance remains a risk through its lack of remote verifiability, as a senior manager explained "*third-party audits, because they are largely paid for by the auditee, can contain inherent conflicts of interest*" [respondent 2].

Information remains as reliable as it was when collected as part of due diligence programs and reported to the relevant parties. The traceability of the information that goes onto the blockchain in terms of legal and regulatory compliance makes firms that join and comply with standards such as RMI more likely to take human rights seriously. These actors potentially

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demonstrate their sustainability credentials by volunteering their information. A senior manager explained that pressure for fuller human rights audits is increasing: "*Assuring against human rights and child labour abuses in these supply chains is one of the big priorities now*" [respondent 1]. The technology does not solve the challenge of guaranteeing working conditions, as a senior manager told us, *"No, our system does not currently do that [human rights assurance],"* adding that this was a *"more complex issue"* [respondent 2].

A residual problem is the nature of some supply chain information. The blockchain solution provides supply chains with a way to demonstrate responsible sourcing through provenance, though the firm is aware that issues remain. In this case example, the Minsur mine produces cassiterite volumes sufficient to act as a single source to meet the downstream smelter and production demands. Employing multiple smaller mining operations would require an initial inspection and audit of each mine to be approved, then onboarding and monitoring to ensure standards are maintained. This increases complexity and incurs incrementally higher system costs with every mine added.

The blockchain system offers reliable traceability of materials to the mine of origin and facilitates the data collection and audit process for human rights in the ION. It does not solve the challenge of reliable data collection of working conditions in and around remote mine sites, which still requires costly, trusted in-person monitoring and audits.

4.3 Enhances efficiency

Our third theme, *Enhances efficiency,* consists of the second-order codes *Inexpensive relative to volumes and values of metals transacted* and *Limits scope for fraud, and helps identify errors*. This theme speaks to the set-up and operational costs of creating a blockchain-based system for ION compliance with mandatory SCT regulation.

Inexpensive relative to volumes and values of metals transacted

Introducing technology-based tools to overcome information processing uncertainties comes with up-front costs. Developing the app and the system around it required investment from a European Union Horizon 2020 grant of €2,300,000 (around $2,600,000). Total costs for the development were €3,300,000 (around $3,750,000). A further grant of €180,000 ($205,000) was awarded by the European Institute of Innovation and Technology (EIT) Raw Materials fund.

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The system employs an energy-efficient proof-of-authority consensus mechanism in so that, "*The power equivalent of only a couple of light bulbs is necessary to run the Minespider blockchain*" [respondent 5]. A senior manager at Minespider confirmed the low costs of operating the system, stating that the total cost of developing and operating the system, including data entry and storage, and conducting transactions, amounts to "*pennies per container*" (respondent 2). At the mid-November 2023 spot price of roughly $24,500/ton, a container holding 25 tonnes of tin is valued at $600,000. Therefore, the system makes for extremely efficient tracing at a tiny fraction of the cost of each container. Minsur mines 12% of the world's tin annually, and in this context, the initial investment in the blockchain traceability system is comparatively small.

Limits scope for fraud and helps identify errors

Allied to the direct financial costs of implementing and managing the system, Minespider's blockchain reduces indirect costs arising from incorrect information entry. The decentralized blockchain in this context proposes two advantages. First, it provides a level of traceability which offers assurance on regulation issues while satisfying customer and consumer demand for information. The ability to allow relevant parties access to information while enabling all firms to control their sensitive information incentivizes supply chain actors to participate. The ION provides the necessary legal and regulatory documentation flows in a system that is controllable and thus useable to both large and small firms alike. The system overcomes the issues of power asymmetry and "*the inherent conflicts between data privacy and transparency in supply chains*" [archival document C].

Second, the immutability of the data held on a blockchain makes the information reliable to the relevant parties while deterring error and fraud from entering the system. Contracting firms know that remote verification of information can expose incorrect information inputs, "*giving us confidence that companies are going to give us data they're willing to stand behind*" [archival document H]. The immutability actually means that more time is available if errors or incorrect data enters the system. As a senior manager explained: "*The question with incorrect data entry is whether incorrect data was entered or whether it has been changed. Well, with a blockchain system, it's extremely hard to change it. But because everything is permanent on a blockchain, we can always find out who entered the data. And we can find that out any time. Only highly authorized persons can put data on the blockchain. And I can check it any time - now or in a year's time. So, the moment to authenticate something*

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is very long. Traditional systems meant that the quicker an error was identified, the higher the chance of identifying how and why this error was made. With a blockchain, there is no rush and the person entering the data onto the system then knows that this can be checked at any time and that they can be identified at any time" [respondent 2].

5. Discussion

Drawing on IPT, this case study contributes to the literature by addressing the ION's use of blockchain to meet mandatory regulatory compliance in the mining industry. We show that blockchain can reduce information uncertainty and equivocality to meet mandatory regulatory compliance in the supply chain. Further, the information processing implications of using blockchain resulted in cost efficiencies and enhanced collaboration in the ION studied.

5.1 Theoretical contributions

In answering our research question, *How does blockchain address information uncertainty and equivocality in assuring regulatory compliance in an interorganizational network?* this study contributes to the IPT literature, specifically with regard to mandatory SCT.

First, we demonstrate how blockchain-based systems can address both environmental uncertainty and equivocality. Environmental uncertainty, arising from the inability to collect sufficient information, is overcome by collaboration and coordination of information flows within the ION facilitated by the blockchain-based system. In the context of the ION, we answer Giri and Manohar's (2023) call for more research on IPT extended beyond an individual firm perspective and on IPT along supply chains. Blockchain-based systems help create trust (Yavaprabhas *et al.,* 2023) required within an ION to ensure partner firms pool information resources to reduce uncertainty (Laari *et al.,* 2023). The system design overcomes the limitations of a reliance on relational aspects of trust (Chowdhury *et al.,* 2023). The system's identification of individuals entering data coupled with the immutability of the data itself means ION members seek to share accurate information. The system moderates ION member's behaviour, maintaining affective trust in supply chains (Brookbanks and Parry, 2022). Our case ION was able to leverage the scale of mining ore from a single, provable source to collect sufficient information on the provenance of the cassiterite mined and to use blockchain as the "single source of truth" posited by Goldsby and Hanisch (2023, p.3) to evidence claims

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transparently. Therefore, our findings demonstrate that prior claims for the need for enhanced cognitive skills (Bode *et al.*, 2011) or broader increases in information capacity (Busse *et al.*, 2017) are not necessarily required when deploying a well-designed blockchain technology-based system.

Blockchain's role as the single source of truth also forms part of our ION case's strategy for managing equivocality. Mining is an industry that suffers from difficulties verifying information (Srinivasan and Swink, 2018). Blockchain reduced equivocality by maintaining records of identity, organization, time, and place, which can be audited at any time. The ability to audit information immediately and at any time in the future limits the impact of errors and reduces the likelihood of fraud. The 'halo' effect that has been claimed, whereby information is believed simply because it is stored on a blockchain (Rogerson and Parry, 2020), therefore, does not hold. The blockchain system reduces ION uncertainty by creating organizational accountability and thus liability for specific data entered rather than establishing data's veracity.

IPT scholars (Galbraith, 1974; Giri and Manohar, 2023; Laari *et al.,* 2023) posit that information uncertainties can be overcome with *increased* information processing capacity. In this research, we find that blockchain offers *different* capabilities. Rather than requiring additional computational power in the form of additional computer or analytical systems, blockchain's disintermediation can overcome information uncertainties by providing information to all relevant parties as that information is recorded. The relationship between the need for more, better information and investment in the processing power required to deliver those enhancements, previously conceptualized as a linear relationship, is overcome by blockchain acting as a bridging mechanism (Manhart *et al.,* 2020), providing a relatively inexpensive method of obtaining and sharing better information. Our study thus builds on the embryonic literature on blockchain and IPT (Galati, 2022; Zhu *et al.,* 2022), which finds that blockchain can enhance information processing efficiencies, adding evidence for the specifics of blockchain's implications for IPT (Aben *et al.,* 2021).

Second, we contribute to the literature on blockchain in OSCM, which also offers managerial implications. We show from our findings that blockchain can offer a method for compliance with SCT for conflict minerals regulation and by extension can provide evidence of provenance required in other regulations. We believe this is the first study empirically demonstrating the use of DPPs in the OSCM literature, confirming the usefulness of DPPs

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conceptualized by Batista *et al.* (2023). Our findings show that the critical mass of adopters Sternberg *et al.* (2021) suggest is necessary for blockchain-based systems to succeed may be over-stated in the context of an ION. Similarly, Baharmand *et al.'s* (2021) claim that private blockchain architectures are unlikely to provide transparency may also be challenged. We confirm Surucu-Balci *et al.'s* (2024) finding that regulation can promote adoption of blockchain. We have demonstrated that a blockchain-based system can provide controlled visibility of information to project parties and external transparency to stakeholders through a layered blockchain, retaining sufficient privacy for parties to trust the system without the system being entirely private. This is one of the first OSCM papers to describe and demonstrate the utility of a multi-layered blockchain system design, building on the work of Hanisch *et al.* (2023), who assert that digital technologies require careful governance. The work responds to calls for empirical research on blockchain (Hastig and Sodhi, 2020; Zhang and Ramesh, 2023; Brau *et al.,* 2024), on overcoming uncertainties using technology (Fosso Wamba *et al.,* 2021), and on SCT (Sodhi and Tang, 2019).

Our identification of the development and operational costs for this specific system is also novel, and an important facet of the discourse on the cost-benefit of blockchain (Pun et al., 2021). Identification of cost relative to volume from our discrete case setting provides some evidence for Lumineau *et al.*'s (2021) claim that blockchain can provide the governance functions necessary for explicit transactions. The blockchain-based system was a trusted source of information and of great importance for transparency and compliance processes. We do not have sufficient evidence to suggest that the system could provide most governance functions, but we provide a starting point for further investigation.

We also identified limitations to blockchain systems. Our findings support Rogerson and Parry's (2020) claim that the physical-digital nexus is a point of weakness for blockchain systems. Though the case study blockchain system offers a reliable record of provenance via digital product passports and for documentation of compliance with human rights legislation, monitoring conditions in the mine and its local community and providing data remains difficult over the longer term. Traditional analogue systems remain in use for the collection of richer data, but reliance on manual data entry retains those risks already documented in the literature (Soundararajan *et al.*, 2021). While blockchain provides the traceability to comply with conflict minerals regulations, the system solution does not currently address the wider challenge of conflict minerals' issues of human rights abuses at mines. Similarly, the immutability of the data in a blockchain creates benefits, but the lack of an editing function (Brookbanks and Parry,

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2022) reduces system efficiency in instances in which records are found to be incorrect, and data issues need to be resolved.

Finally, our findings highlight that a small network of firms can benefit from a well-designed blockchain system. Goldsby and Hanisch (2022) posit that poor governance design is responsible for the failure of many blockchain adoptions. We highlight potential paradoxes in this respect. Sternberg *et al.* (2021) suggest that blockchains require scale to fulfil their potential. We problematize this claim with our case study, finding that a small ION can benefit from blockchain where the network can retain governance mechanisms and align priorities, but that may not hold in a larger ION where varying priorities create tensions. Similarly, the small group of firms involved in Minespider's blockchain and its decentralized governance avoids the tensions that centralization can create in blockchain systems (Naef *et al.*, 2022).

5.2 Managerial implications

Our study highlights important considerations for managers seeking to use blockchain for SCT through IONs. We show that there are benefits and limitations of blockchain, which need to be carefully considered. We demonstrate that, in the right use case, blockchain can be solve some of the issues that create intractable regulatory issues and do so at a tiny fraction of the revenue from the material or product being produced. Blockchain can efficiently and affordably enable the collection of information to assure the provenance of materials where prior solutions had left transparency gaps. By using blockchain as part of an ION, managers can collect required information and ensure that the information is more reliable. That information is not deemed true because it is stored on a blockchain, but the blockchain provides a single source of information auditable at any time after information input, linking relevant parties' identities to information inputs. The ability to hold parties liable for their claims is a key finding for managers. Similarly important is the fact that blockchain solutions studied here and in the empirical literature have not yet integrated technologies sufficiently to collect all the richer sustainability information demanded by stakeholders and regulators. Blockchain-based systems can, therefore, demonstrably provide assurance of provenance for compliance with mandatory SCT under conflict minerals regulation. We anticipate that blockchain will likely be similarly useful in materials' supply chains with similar characteristics to those in the mining industry. However, while such qualities will be a useful starting point for the due diligence expected of forthcoming regulation, current blockchain systems will continue to rely on

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traditional in-person auditing to execute on-site and community checks of human rights and environmental conditions.

5.3 Further research

Weaknesses remain in the continued use of blockchain to collect only basic certification information rather than information that might offer better evidence of human rights considerations. Publicly available information that allows customers and consumers to check for themselves what actions are taken in and around the mine might enable wider trust. Creating a traceability system for compliance that deals somewhat opaquely with issues of social auditing could itself create trust issues. Compliance and social audit-led actions are known to suffer from a lack of worker-centricity (Soundararajan *et al.,* 2021), even as voice-of-worker methods of information collection have been extensively tested. We, therefore, identify a boundary condition of blockchain's traceability potential, since the human rights information underpinning the compliance enabled by blockchain systems remain opaque. Future research should investigate how such information can be collected and how a system might be scaled to assure minerals from multiple sites, with the governance challenges involving more parties may bring. Future research should also seek to understand the differences between using blockchain to trace conflict minerals and blockchain and other traceability tools' applicability to wider mandated traceability and transparency regulation under the new EU Corporate Sustainability Due Diligence legislation, which the European Parliament passed in December 2023 (European Council, 2023).

6. Conclusion

Information processing theory has been employed to explain various supply chain management findings (Giri and Manohar, 2023). Through our case study context of the ION, we answer Giri and Manohar's (2023) call for more research on IPT extended beyond an individual firm perspective and on IPT along supply chains. IPT has only recently been used to explore blockchain use in supply chain contexts (Zhu *et al.,* 2022), and we add to this body of work. Studies exploring IPT and blockchain show the capability of blockchain to increase processing capability (Galati, 2022), and this paper contributes to knowledge by showing that a blockchain-based system can address two of the key IPT challenges: environmental uncertainty and equivocality. We further contribute through the identification of some of the limitations of a blockchain system, which have received much less attention in the literature. By enabling the

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collection and analysis of increased amounts of information without proportional increases in processing capacity, materials can be traced through supply networks to provide regulatory compliance. The case example blockchain system for tin supply chains does so at a tiny fraction of one percent of the value of the materials being traced. Current systems lack the ability to reliably collect richer data on human rights and environmental conditions, which are part of conflict minerals considerations.

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Appendices

Appendix A – List of data sources and their use in the data collection and analysis

Appendix B – List of primary data sources

Appendix C – List of secondary data sources cited in this study

Appendix D – List of secondary data sources

Appendix E – Interview protocol

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Appendix A – List of data sources and their use in the data collection and analysis

Data source	Type of data	Use in the analysis
Semi-structured interviews	**First round – Pilot interviews (2019)** Interviews with informants in the mining industry, including Senior Managers and Innovation Managers.	Pilot interviews: Testing/confirming the appropriateness of the case, testing the suitability of interview questions and refining the interview questionnaire. Gathering data regarding the industry characteristics and supply chain blockchain challenges in this industry. Gaining an initial understanding of the adequacy of blockchain to combat information uncertainty and information equivocality.
	Second round (2019 – 2022) Interviews and email exchanges with informants in the mining industry, including Senior Managers and Innovation Managers. The total number of primary data: 22 hours of verbal interviews and 42 emails	Expanding the sample to evaluate blockchain efficacy to combat information uncertainty and information equivocality. Gathering data and clarifying specific points to understand disparities between secondary and interview data. Presentation and verification of the findings and conclusion of this study.

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Secondary data sources	**Publicly available documentation (2019–2022)** Printed/online: Documents about the industry (e.g. companies involved and industry reports), firm disclosures, social media posts, social media videos and whitepapers. The overall number of secondary data: 390 pages and ~1.15 hours video analysis.	Gaining an initial understanding of the mining industry and supply chain transparency challenges, insights into industry practices using blockchain and a higher-level understanding of the uncertainties in this industry and future trends. Supporting, integrating, and cross-checking interviewee-references to enhance data validity. Gathering data on blockchain efficacy to address information uncertainty.

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Appendix B – List of primary data sources

No.	Interviews	Video Time	Details
1	Respondents 1 & 2	2:30:00	Initial, exploratory discussions
2	Respondents 1 & 2	2:30:00	Initial, exploratory discussions
3	Respondent 1	00:45:57	Discussion on project detail
4	Respondents 2 & 3	00:39:03	Discussion on technical detail
5	Respondent 1	01:25:40	Discussion on supply chain factors
6	Respondent 2	01:49:23	Discussion on supply chain factors
7	Respondent 4	00:55:03	Discussion on project detail
8	Respondent 3	00:27:10	Discussion on technical detail
9	Respondent 4	00:52:09	Product passport walkthrough
10	Respondent 5	00:49:03	Discussion on technical detail
11	Respondent 4	00:58:52	Discussion on project and product integration
12	Respondent 2	01:56:12	Checking of details
13	Respondent 2	02:30:00	Clarification of data
14	Respondent 2	02:30:00	Agreeing final paper contents
15	Respondents 1& 2	2:00:00	Follow-up calls to discuss scalability and similar issues pertaining to future roll-out of the system
Total duration of interviews: 22 hours			

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No.	Email exchanges	No.	Details
16	Respondents 1 & 2	42 pages	Over a period of 18-months, the second author proactively interacted with the Minespider managers.
Total number of email exchanges: **42 pages**			

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Appendix C – List of secondary data sources cited in this study

Document title	Document ref.	Source
Supply chain meets blockchain for end-to-end mineral tracking	A	https://sustainability.google/progress/projects/traceability/
Protocol for Due Diligence in the Raw Material Supply Chain	B	https://uploads-ssl.webflow.com/5bb20121ca2e96ee01db29bc/5c0fa81d4a4585e37ea764b7_Minespider_Whitepaper.pdf
How mining supply chains can benefit from blockchain	C	https://www.industrialsage.com/nathan-williams-of-minespider-on-blockchain/
Book Chapter by Minespider CEO and Regulatory Specialist: A Transparent New World: Ethically Sourced Mineral Supply Chain Contributors: By Nathan Williams and David Lee Williams	D	Williams, N., Williams, D.L. and Dutta, S.K., 2020. A Transparent New World: Ethically Sourced Mineral Supply Chain. In The Definitive Guide to Blockchain for Accounting and Business: Understanding the Revolutionary Technology. Emerald Publishing Limited. Pg. 149-208.
Nathan Williams, Minespider Facebook video 18/2/21	E	https://www.facebook.com/watch/?v=989442261583556
From mine to factory: Volkswagen makes supply chain transparent with blockchain	F	https://www.volkswagen-newsroom.com/en/press-releases/from-mine-to-factory-volkswagen-makes-supply-chain-transparent-with-blockchain-4883/download
Data-driven, data-dependent: Blockchain in the Mining Industry	G	https://www.minespider.com/blog/data-driven-data-dependent-blockchain-in-the-mining-industry

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Appendix D – List of secondary data sources

No.	Type of data:	Time/Pages	Link
Video Archive (Explanation of technology/recorded interviews/conference presentations)			
1	Facebook Video | How do you balance data privacy and transparency in supply chains?	00:04:53 hrs	https://fb.watch/9XdZulfGvI/
2	Youtube Video | Minespider Endless use cases: The problems we still have to solve in the raw materials supply chain	00:18:45 hrs	https://www.youtube.com/watch?v=tNJ7JixGs1w
3	Facebook Video | What are the latest developments in conflict mineral regulation?	00:05:11 hrs	https://fb.watch/9XdQdCtgwj/
4	Facebook Video | The Tin Traceability Project	00:05:36 hrs	https://fb.watch/9XdWHowRRd/
5	Youtube Video| Minespider TV Episode 1	00:20:05 hrs	https://www.youtube.com/watch?v=pjRRXc1jlAA
6	Youtube Video | Minespider TV Ep 2 | Tech Talks: Layers	00:08:27 hrs	https://www.youtube.com/watch?v=w4OVAA9rLeQ
7	Youtube Video | Minespider TV Ep3 | How can blockchain improve trust and efficiency for companies?	00:03:19 hrs	https://www.youtube.com/watch?v=vEH08RZB9IA
8	Youtube Video | Minespider TV Ep5 | How can we make blockchain interoperability work?	00:04:00 hrs	https://www.youtube.com/watch?v=Ilao-MVJe8c

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9	Youtube Video | Minespider in 30 secs	00:00:30 hrs	https://www.youtube.com/watch?v=SUx9FqRngic
10	Youtube Video | Minespider TV Ep6 | How do you manage data permissions without a central party?	00:02:38 hrs	https://www.youtube.com/watch?v=rXCWraw5paY
11	Youtube Video | Minespider TV Ep7 | What are the latest developments in conflict mineral regulation?	00:05:11 hrs	https://www.youtube.com/watch?v=aT9pKQm5EFA
12	Leading tin producer LuNa Smelter tracking on Minespider's blockchain	00:04:05 hrs	https://www.youtube.com/watch?v=HEQ5aeoiyPw
Internal: ***Company Website/ Blog/ Reports***			
13	Minespider | Medium.com | The Scramble to rare earth elements	8 pages	https://medium.com/minespider/the-scramble-to-secure-rare-earth-elements-e2606d44ce09
14	Minespider | Medium.com | Evolving legislation from Dodd-Frank to the EU Conflict Minerals Regulation	6 pages	https://medium.com/minespider/evolving-legislation-from-dodd-frank-to-the-eu-conflict-minerals-regulation-751289d562a4
15	Minespider | Medium.com | Data-driven, data-dependent: Blockchain in the Mining Industry	6 pages	https://medium.com/minespider/data-driven-data-dependent-blockchain-in-the-mining-industry-bfe3c7709c6

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16	Minespider | Medium.com | Is blockchain on your to-do list? It should be.	5 pages	https://medium.com/minespider/is-blockchain-on-your-to-do-list-it-should-be-84e9b143e523
17	Minespider | Medium.com | The false hope of blockchain interoperability.	8 pages	https://medium.com/minespider/the-false-hope-of-blockchain-interoperability-258da2af62b
18	Minespider | Medium.com | Tokenizing the mineral supply chain.	7 pages	https://medium.com/minespider/tokenizing-the-mineral-supply-chain-2eef680f2e44
19	Minespider | Medium.com | What are conflict minerals	4 pages	https://medium.com/minespider/what-are-conflict-minerals-28d66b0a1279
20	Minespider | Medium.com | Conflict Mineral Regulations.	6 pages	https://medium.com/minespider/conflict-mineral-regulations-c0058a9f929d
21	Minespider | Upstream producers	1 page	https://www.minespider.com/upstream-producers
22	Minespider Facebook page | QR code image, 7/6/21	1 page	https://www.facebook.com/Minespider/photos/a.266112284228866/960030554837032/
23	MDPI Proceeding Paper authored by Minespider Managers at Minespider	12 pages	Bacchetta, A.v.B., V. Krümpel, E. Cullen. 2021. Transparency with Blockchain and Physical

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			Tracking Technologies: Enabling Traceability in Raw Material Supply Chains. *Material Proceedings*, 5 (1). https://www.mdpi.com/2673-4605/5/1/1
24	Minespider White Paper | Protocol for Due Diligence in the Raw Material Supply Chain	29 pages	https://uploads-ssl.webflow.com/5bb20121ca2e96ee01db29bc/5c0fa81d4a4585e37ea764b7_Minespider_Whitepaper.pdf
External: ***News articles/ Case Studies/ Reports written by others***			
25	Statecraft Tech | Medium.com | Minespider partners with Google for tin blockchain traceability	3 pages	https://medium.com/ktrade/minespider-partners-with-google-for-tin-blockchain-traceability-d08e72630d7f
26	Google Sustainability Supply chain meets blockchain for end-to-end mineral tracking	5 pages	https://sustainability.google/progress/projects/traceability/
27	Ledger Insights | Minespider partners with Google for tin blockchain traceability	3 pages	https://www.ledgerinsights.com/minespider-partners-with-google-for-tin-blockchain-traceability/
28	Fairtin	1 page	https://www.fairtin.com

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29	RMI Responsible Minerals Assurance Process: Tin and Tantalum Standard	2 pages	http://www.responsiblemineralsinitiative.org/about/rmi-initiative
30	RMI Responsible Minerals Assurance Process: ESG Standard for Mineral Supply Chains	2 pages	http://www.responsiblemineralsinitiative.org/responsible-minerals-assurance-process/
31	Conflict Minerals Reporting Template (CMRT)	2 pages	http://www.responsiblemineralsinitiative.org/reporting-templates/cmrt/
32	Volkswagen | From mine to factory: Volkswagen makes supply chain transparent with blockchain	3 pages	https://www.volkswagen-newsroom.com/en/press-releases/from-mine-to-factory-volkswagen-makes-supply-chain-transparent-with-blockchain-4883/download
33	Cordis | Horizon 2020 Blockchain Protocol for Responsible Mineral Sourcing, Minespider Grant	3 pages	https://cordis.europa.eu/project/id/946437
34	Google | Responsible Supply Chain Report 2019	40 pages	https://services.google.com/fh/files/misc/google_2019-rsc-report.pdf
35	Electronics Manufacturing by Petra Gottwald [translated from German language to English] | How important is sustainability when it comes to soldering?	12 pages	https://www.all-electronics.de/elektronik-fertigung/welchen-stellenwert-hat-nachhaltigkeit-beim-

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			thema-loetzinn-389.html?fbclid=IwAR2cudWZspH5vuVlBW9h6PCfo0ETSbZoBV_bv9-lvkQerlN6Bp69Is3hyB8
36	Global Mining Review | Minery to integrate Minespider blockchain traceability	1 page	https://www.globalminingreview.com/finance-business/15102021/minery-to-integrate-minespider-blockchain-traceability/
37	Minsur Sustainability Report 2019	47 pages	https://www.minsur.com/wp-content/uploads/pdf/Memoria%20Anual/ENG/2019%20Minsur%20Sustainability%20Report.pdf
38	International Mining IM | Minespider looks to scale blockchain platform following fundraising, 16/4/2020	1 page	https://im-mining.com/2020/04/16/minespider-looks-scale-blockchain-platform-following-fundraising/
39	Sifted by Pia Behme| Startups are helping big companies build ethical supply chains – but is the tech ready? | 15/10/2019	5 pages	https://sifted.eu/articles/minespider-ethical-supply-chain/
40	TechRepublic| How Minespider uses blockchain to ensure an ethical supply chain | 19 Apr 2019	6 pages	https://www.techrepublic.com/article/how-minespider-uses-

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			blockchain-to-ensure-an-ethical-supply-chain/
41	Mining Review Africa | How mining supply chains can benefit from blockchain | 10 Dec 2020	3 pages	https://www.miningreview.com/battery-metals/how-mining-supply-chains-can-benefit-from-blockchain/
42	Mining Review Africa | Minespider partners with LuNa Smelter to pilot blockchain tool in Rwanda | 18 Nov 2020	2 pages	https://www.miningreview.com/east-africa/minespider-partners-with-luna-smelter-to-pilot-blockchain-tool-in-rwanda/
43	Mining Review Africa | Minespider to develop a responsible minerals compliance tool | 15 Sept 2020	2 pages	https://www.miningreview.com/battery-metals/minespider-to-develop-a-responsible-minerals-compliance-tool/
44	International Mining | Volkswagen signs up blockchain specialist Minespider to track | 25 Apr 2019	3 pages	https://im-mining.com/2019/04/25/volkswagen-signs-up-blockchain-specialist-minespider-to-track-lead-supply/
45	The New Times | RwandaNew partnership to ensure transparency in mineral supply chain | 21 Nov 2020	3 pages	https://www.newtimes.co.rw/news/new-partnership-ensure-transparency-mineral-supply-chain

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46	Mining Magazine | Rwanda smelter to pilot OreSource blockchain tool | 20 Nov 2020	2 pages	https://www.miningmagazine.com/supply-chain-management/news/1399730/rwanda-smelter-to-pilot-oresource-blockchain-tool
47	Raconteur | Cobalt mining in the DRC: the dark side of a clean future | 4 Jun 2019	1 page	https://www.raconteur.net/corporate-social-responsibility/cobalt-mining-human-rights/
48	Orbis Research - Rendering Conscientious Research | Global Blockchain in Metals and Mining Market 2021 Research Report by User Demand, Consumption, Applications, Types, Trends and Forecast 2026 | Jan 02 2021	8 pages	https://www.globenewswire.com/news-release/2021/05/26/2236128/0/en/Global-Blockchain-in-Metals-and-Mining-Market-2021-Research-Report-by-User-Demand-Consumption-Applications-Types-Trends-and-Forecast-2026.html
49	Automotive Logistics | Blockchain: Potential unlocked? | 4 Jul 2019	9 pages	https://www.automotivelogistics.media/supply-chain-management/blockchain-potential-unlocked/38624.article
50	CIM Magazine |	7 pages	https://magazine.cim.org/en/news/2019/former-barrick-cio-michelle-ash-

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	Former Barrick CIO Michelle Ash to join blockchain start-up | 25 Jan 2019		to-join-blockchain-start-up/
51	Mining.com | How blockchain can track conflict minerals | 25 Sept 2019	8 pages	https://www.mining.com/how-blockchain-can-track-conflict-minerals/
52	Volkswagen Sustainability Report 2019 | 17 Mar 2020	100 pages	https://www.volkswagenag.com/presence/nachhaltigkeit/documents/sustainability-report/2019/Nonfinancial_Report_2019_e.pdf
53	Northwest Diamond Notes | Blockchain in Metals and Mining Market by Technology ...| 13 Dec 2021	4 pages	https://www.nwdiamondnotes.com/blockchain-in-metals-and-mining-market-97271/
54	We're thrilled to announce that Minespider raised has €2.8 million! We want to say a huge thank you to everyone who has supported us, especially our friends at @VWGroup_DE, @Google, @Cisco, @SGS_SA, Stannol, & Minsur. https://minespider.com/press #startup #funding #blockchain #mining	1 page	https://twitter.com/MinespiderTeam/status/1250755887003729925
55	We made in on to @WDR! 🖥 Minespider tracks tin as it's responsibly sourced from @Minsur's San Rafael mine in #Peru, to #Stannol's fair #tin solder, & as it eventually makes its way into	1 page	https://twitter.com/MinespiderTeam/status/1326867414156447746

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	@VWGroup's vehicles: https://www1.wdr.de/mediathek/video/sendungen/lokalzeit-bergisches-land/video-loetmittelhersteller-bietet-fair-gehandelten-zinn-an-100.html #WDR #Deutschland #mining #sourcing #blockchain		
56	Are you ready to comply with #Germany's new #SupplyChainDueDiligenceAct? The aim is to protect #humanrights and the #environment in a global economy by obliging #German companies of a certain size to perform better #duediligence on their #supplychains. https://bit.ly/3yrsw2L	1 page	https://twitter.com/MinespiderTeam/status/1423309396986187779
57	Ever wondered how a #blockchain project is implemented? Check out how we set up Minespider's #traceability software on the ground at @LuNa_Smelter in Rwanda. #usecase #Rwanda #mining #metals #tin #video #responsiblemining #responsiblesourcing #ICGLR	1 page	https://twitter.com/MinespiderTeam/status/1408078243492663299
58	Advantages of #blockchain in #mining and #metals: beyond supply chains Christian Ecker from	1 page	https://twitter.com/EEIPDigital/status/1403321225003347978

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	@MinespiderTeam: use cases from data management to property registration, contract mgmt and finance https://buff.ly/3vkk8Qg @kamni_chain #BlockchainforEU #digitalisation #sustainability		
59	With the #EU #ConflictMineral #Regulation now in full force, our founder, @NWilliamsMBA, reflected on the shifts taking place in the regulatory environment for #mineral #sourcing. #responsiblesourcing #education #compliance #traceability #transparency	1 page	https://twitter.com/MinespiderTeam/status/1356956735819440128
60	"Companies may now be incentivised to be #responsible and #transparent as part of the cost of doing business," @NWilliamsMBA. Read the full digital copy of our feature in @MiningReview Issue 11, 2020! https://miningreview.com/battery-metals/how-mining-supply-chains-can-benefit-from-blockchain/ #blockchain #rawmaterials #mining #traceability	1 page	https://twitter.com/MinespiderTeam/status/1339198263300399107

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61	What are some solutions for the #sustainable #production of #rawmaterials? Join the @EU_EESC's '#Blockchain technologies for the #Mining Industry' event and hear from our founder, @NWilliamsMBA, about how these technologies work in practice. https://eesc.europa.eu/en/agenda/our-events/events/blockchain-technologies-mining-industry/programme	1 page	https://twitter.com/MinespiderTeam/status/1338457582021865475
62	Have a listen in to the latest episode of @MiningReview's #Modernisation in #Mining #podcast! □ Senior Deputy Editor Chantelle Kotze spoke to our founder and CEO, @NWilliamsMBA, about #blockchain for the #mining sector. https://miningreview.com/multimedia/podcasts/deep-insights-24-modernisation-in-mining-ep-5/ #miningindustry #miningnews #Africa	1 page	https://twitter.com/MinespiderTeam/status/1336245166542376965
Total secondary material: **~390 pages, ~1.15 hours video material**			

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Appendix E – Interview protocol

Initial introductory questions to establish:

- Respondent position
- Respondent role
- Respondent experience

What are the information demands underpinning the Minespider blockchain?

What are the information problems that the Minespider blockchain seeks to solve?

How does the Minespider blockchain address these problems?

How has the Minespider blockchain enabled the different firms in the network to work together?

What are the costs involved in setting up and running the Minespider blockchain?

What problems in this sphere does the Minespider blockchain not currently address that you would like it to?

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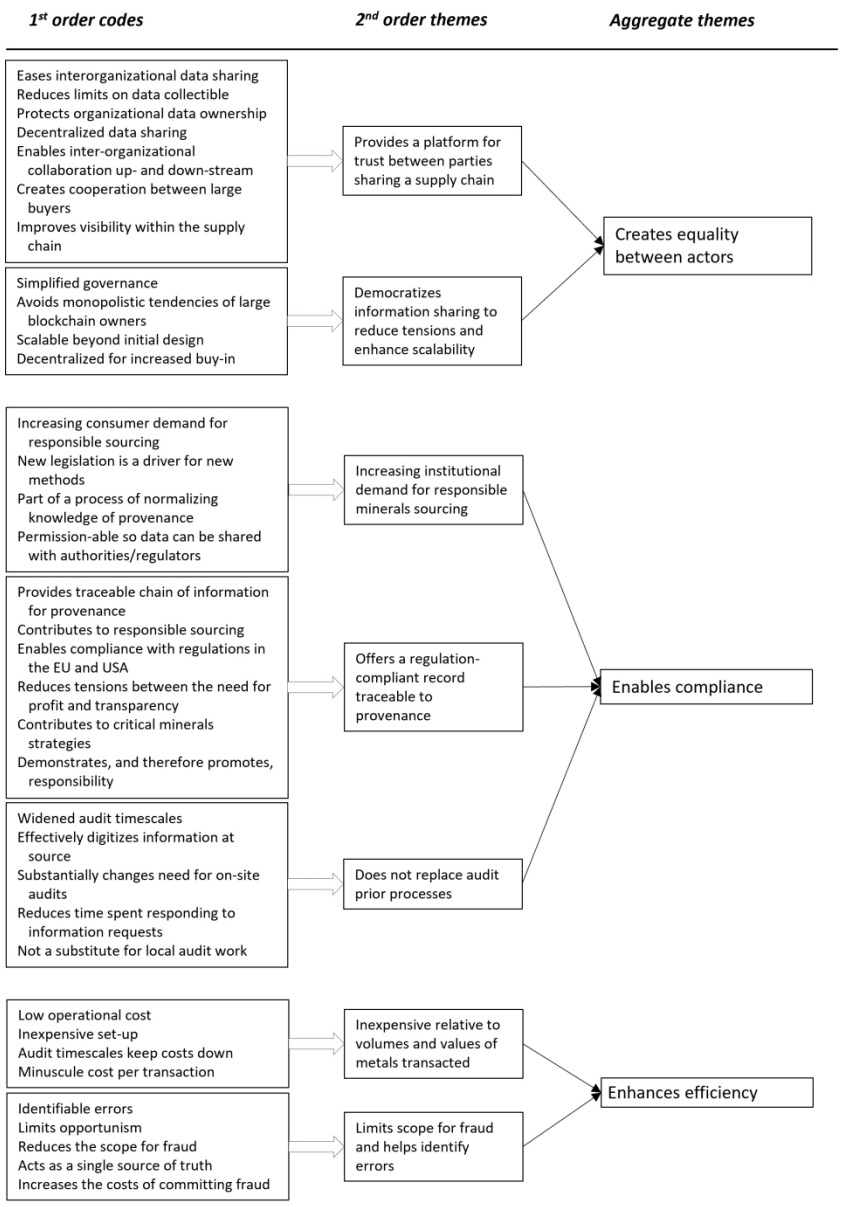


Figure I: Data analysis process

186x268mm (330 x 330 DPI)

Table I: Methodological framework

Research question: *How does blockchain address information uncertainty and equivocality in assuring regulatory compliance in an interorganizational network?*

Research approach, sampling and setting	Qualitative: Exploratory case study Sampling criteria: (i) Use of blockchain for regulatory compliance (ii) A complex dispersed network of organisations who are interdependent with respect to data integrity, such that blockchain technology enables trusted data provision Industry: An industry with high information processing need to fulfil compliance regulations and an industry that encounters major supply chain transparency information (i.e. mining industry). Firm selection: Access to inter-organizational network (ION) in the mining industry. Out of four industry blockchain service providers only one was able to provide information about ION's supply chain transparency challenges and need for information processing (i.e. Minespider).
Data collection & sources	Interviews and secondary data sources (2019-2022) (i) Pilot interviews and initial secondary data collection (ii) Semi-structured interviews (iii) Secondary data sources (i.e. industry reports, company website data, press releases, media articles, social media)
Analysis, interpretation and reporting	Open, axial and selective coding (Strauss and Corbin, 1990). Ensuring intercoder reliability through comparison of codes and discussions among researchers to ensure comparability. Verification of research results through interviewees.

Table II: Second-order themes and supporting quotes

CREATES EQUALITY BETWEEN ACTORS		
Coding	Supporting Quotes	Source
Provides a platform for trust between parties sharing a supply chain	"It's this nested technology that makes it so that you are in total control of your data, but there's no bank, or no central authority, or no central company, or government that has backdoor access to it."	Respondent 1
	"Responsible producers are often at a disadvantage in the global market."	Respondent 1
	"Avoids the risk of a large player with a private blockchain creating a monopoly on the global mineral supply chain. Minespider's design provides a way to decentralize protocol governance while preserving the privacy of proprietary data."	Company report (archival document A)
	"Competing companies often wish to use their own system to avoid having their supply chain data visible to competitors, resulting in multiple competing systems that are not interoperable."	Minespider whitepaper (archival document B)
	"Supply chain data must be self-sovereign. Neither Minespider nor other actors on the platform should be able to access supply chain data they do not own."	Minespider whitepaper (archival document B)
	[Minespider's blockchain gives] us confidence that companies are going to give us data they're willing to stand behind."	Social media post (archival document E)
Democratizes information sharing to reduce tensions and enhance scalability	"These layers divide the data into a private layer, offering certainty for regulation; a transparency layer, sharing data downstream for certifications and provenance, but not competitors or the public; and a public layer available for anyone to see. The key to this system is that the encryption is done by the data owner, meaning that they alone have the power over those data. This is how our blockchain is decentralized, removing the need for a central authority holding passwords. It's a system designed without a central authority."	Company news release (archival document F)
	"Due to the sensitive nature of supply chain data, most… pilots have been developed on private permissioned blockchains run by a single end user."	Minespider whitepaper (archival document B)
	"It runs without Minespider. We add nodes when needed, but the system runs without us."	Respondent 2

	"Many suppliers provide visibility into the sources of their raw materials through spreadsheets—namely the Responsible Mineral Initiative's Conflict Minerals Reporting Template—that are designed to aggregate information from all the upstream vendors. Suppliers then import these spreadsheets into various tools that require constant calibration, with limited automation."	Respondent 2
	"Our protocol is public and decentralized. It's designed to transform the entire raw materials industry so that dark money and illicit practices become a shadow of the past."	Respondent 2
	"Data collected will be stored as encrypted self-sovereign data packet "certificates", under complete control of the data owner."	Minespider whitepaper (archival document B)
ENABLES COMPLIANCE		
Coding	Supporting quotes	Source
Increased institutional pressure for responsible minerals sourcing	"We need to ensure countries and companies can access critical raw materials and there's increasing demand to know there's no dirty labour in those supply chains."	Respondent 1
	"It is therefore Minespider's plan that if you provide customers with the ability to know where their raw materials come from, suddenly you start to level the playing field and the companies that do not provide data are now at a disadvantage."	Industry news (archival document C)
	"Knowing where your metals come from that go into electronics is becoming normal, expected."	Respondent 2
	"Consumers are demanding more and more from [the minerals their products] contain."	Respondent 2
	"There are constant requirements for information [and that the original proposal was] to see if blockchain could help us reduce the volume of requests and responses."	Respondent 1
	"Assuring against human rights and child labour abuses in these supply chains is one of the big priorities now."	Respondent 1

Offers a regulation-compliant record traceable to provenance	"Like most mined minerals, tin has been historically challenging to track as it travels from its original source through the transportation process to smelters, producers, and finally customers across the world. Its multi-step journey is often opaque."	Company report (archival document A)
	"The entire regulatory environment has been moving more and more towards conducting due diligence. The EU has stepped up regulation to stem flows of resources from the sale of metals into armed conflicts, where human rights abuses occur in and around supply chains."	Respondent 1
	"Opacity gives licence to illicit actors to commit human rights abuses and go undetected."	Respondent 2
	"The mining industry right now is facing the same scrutiny that the textile industry experienced more than a decade ago, when human rights issues first surfaced in sweatshops. Except back then, the textile industry did not have the technology to conduct end-to-end supply chain tracking."	Respondent 1
	"Minespider's position is that an open system is best, and that the issues of data quality assurance may be dealt with by incorporating the possibility of independent data quality audits and evaluation. Multi-stakeholder groups can then create standards for appropriate data and events to be captured by the protocol and use existing certifications and document costs to assign value to the data assets."	Minespider whitepaper (archival document B)
	"Our system gives companies insight into the origin of the metals and the conditions under which they have been produced."	Respondent 1
Does not replace audit processes	'Product passports', described as, "a blockchain-secured digital identification for a material shipment. It contains all the information needed to cross a border, comply with due diligence requirements, and recycle your minerals"	Social media post (archival document E)
	"As tin is mined, a product passport is created for that tin, which is combined with other product passported tin until a shipping container of tin is	Respondent 4

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	available, at which point a new passport is created, subsuming the prior passports within it."	
	"The question with incorrect data entry is whether incorrect data was entered or whether it has been changed. Well, with a blockchain system, it's extremely hard to change it. But because everything is permanent on a blockchain, we can always find out who entered the data. And we can find that out any time. Only highly authorized persons can put data on the blockchain. And I can check it any time - now or in a year's time. So, the moment to authenticate something is very long."	Respondent 2
	"Third-party audits, because they are largely paid for by the auditee, can contain inherent conflicts of interest."	Book chapter (archival document D)
	"A record is collected for the volume mined in a day, which goes into the product passport, and then those records become sub-records one sufficient material has been collected for a container of tin."	Respondent 4
ENHANCES EFFICIENCY		
Coding	Supporting quotes	Source
Inexpensive relative to volumes and transaction values	"The power equivalent of only a couple of light bulbs is necessary to run the Minespider blockchain."	Respondent 5
	"The cost of gathering due diligence data has fallen on the miners in poorer regions. This creates a negative incentive for sourcing legally, as these miners receive the world market price for their minerals, while having to incur increased costs."	Minespider whitepaper (archival document B)
	"The costs of responsible sourcing act as a negative incentive for small scale producers to participate."	Minespider whitepaper (archival document B)
	"The total cost of developing and operating the system, including data entry and storage, and conducting transactions, amounts to pennies per container."	Respondent 2
	"As the number of brands using private blockchains increase, upstream suppliers may find themselves working with 20 or 30 different blockchain systems	Minespider whitepaper

	that all function differently, do not communicate with each other, and have different features and functions. This adds an enormous organizational cost to the upstream and can result in error if the systems are neglected."	(archival document B)
Limits scope for fraud and helps identify errors	"Lack of visibility doesn't mean lack of interest. In fact, as awareness of human rights and ecological issues rises, consumers are demanding more and more from their products—and the minerals they contain."	Respondent 1
	"Traditional systems meant that the quicker an error was identified, the higher the chance of identifying how and why this error was made. With a blockchain, there is no rush and the person entering the data onto the system then knows that this can be checked at any time and that they can be identified at any time."	Respondent 2
	"This system builds more trust into the system by putting more pressure on the person who enters the data. […] When you know the data goes onto a blockchain, you behave differently [...] In the physical world we can do more and better because we can check at any time after the fact."	Respondent 2
	"Optimizing supply chains, eliminating sources of error, guaranteeing social and ecological standards – these are the goals of the collaboration between the Volkswagen Group and Minespider."	Company news release (archival document F)
	"The data being immutable can mean it is auditable. Larger scale fraud may be able to be detected in the long run because data on how much material was shipped would not stand up if the auditors were rotated."	Minespider whitepaper (archival document B)

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